

Functional forms and interactions

Slide route: Lecture7v2.tex: polynomials, logarithms, and interactions through wrapping up. Chapter 8.

Session 9 | A slope need not be constant

Outcomes: Interpret quadratic terms, distinguish log specifications, and calculate a conditional effect in an interaction model.

100-minute route

0–10	Retrieve coefficient units and joint tests.
10–30	Nonlinear functions and polynomials.
30–50	Linear-log, log-linear, log-log examples.
50–55	Pause.
55–80	Dummy/continuous and continuous/continuous interactions.
80–90	Integrate interpretation with tests.
90–100	Exit exercise and Midterm 2 preparation.

Worked example | Quadratic experience

Use with slides Lecture7v2 slides 8–13: Nonlinear Functions and Polynomials in X . Use the derivative and turning-point calculation on slides 9–13; pause before moving to logs.

For the synthetic fitted relationship

$$\widehat{wage} = 10 + 2x - .05x^2, \quad \frac{\partial \widehat{wage}}{\partial x} = 2 - .1x,$$

the marginal effect is 1 at $x = 10$ and 0 at $x = 20$. The turning point is 20. The exact fitted change from 10 to 11 is

$$2 - .05(11^2 - 10^2) = .95,$$

not exactly the derivative at 10. Only discuss the turning point substantively if it lies within a relevant supported range.

Board comment The model is nonlinear in x but linear in the coefficients. OLS can estimate it after including both x and x^2 .

Log interpretation crib

Model	Interpretation of β
$Y = \alpha + \beta \log X + u$	1% higher X : approximately $.01\beta$ units higher Y .
$\log Y = \alpha + \beta X + u$	One unit higher X : approximately $100\beta\%$ higher Y .
$\log Y = \alpha + \beta \log X + u$	β is the elasticity of the modeled log relationship.

Worked examples | Logs and heterogeneous effects

Use with slides Lecture7v2 slides 14–21 for the log examples and slides 22–39 for interactions. Use the log-outcome calculation on slides 18–21, then use the program-by-experience example on slides 26–38.

If a log-outcome model has coefficient .08, a one-unit increase changes the fitted log index by .08. The corresponding multiplicative change is

$$100(e^{.08} - 1) \simeq 8.33\%,$$

versus the 8% approximation. Exponentiating a fitted conditional log mean does not automatically give the conditional arithmetic mean of Y ; retransformation requires attention to the error distribution.

For a level-log model with coefficient 20, a 10% increase in X implies $20 \log(1.10) \simeq 1.906$ outcome units, compared with the approximation 2.

Interaction board example

Let D be a program indicator and x be experience:

$$\hat{Y} = 10 + 2x + 3D + 1.5(xD).$$

For $D = 0$, the line is $10 + 2x$; for $D = 1$, it is $13 + 3.5x$. The program-group contrast at experience x is $3 + 1.5x$, so it equals 9 at $x = 4$.

Ask Is 3 the program effect for everyone? No, it is the contrast at $x = 0$. Is 1.5 the entire program effect? No, it is the difference in slopes. A causal reading still needs identification assumptions.

Test at $x = 4$ Test $\beta_D + 4\beta_{xD} = 0$, using the covariance of the two estimates. Testing only the main dummy coefficient does not answer this question.

Continuous interaction For $Y = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 XZ + u$, the X derivative is $\beta_1 + \beta_3 Z$. State the value of Z whenever reporting it.

Exit answers A log-log slope of $-.7$ implies roughly a .7% decrease for a 1% increase in X . A quadratic model can be estimated by OLS. An interaction coefficient is not a universal total effect.

If short Use one example of each functional form and one interaction in depth; assign repetitive interaction plots for self-study. Proposed Midterm 2 coverage is Lectures 5–7, using prerequisites from Lectures 1–4.

Source: Lecture7v2.tex; synthetic numerical illustrations.